

CIVIL ENGINEERING

Overview

Civil engineers apply principles of motion and materials to plan, design, construct, maintain, and operate public and private facilities, while working under economic, social, and environmental constraints. Commonly included are transportation systems; bridges and buildings; water supply, pollution control, waste management, irrigation, and drainage systems; river and harbor improvements; dams and reservoirs. Civil engineering also includes planning, designing, and executing surveying operations and locating, delimiting, and delineating physical and cultural features on the earth's surface. Research, testing, sales, management, and related functions are also a part of civil engineering. Work on campus is supplemented by inspection trips, which furnish an opportunity for firsthand study of engineering systems in operation, as well as projects under construction.

Environmental engineering, as an emphasis in civil engineering, is concerned with protecting the public and natural health; providing an ample safe water supply; managing solid and hazardous waste; treating and disposing of domestic and industrial wastewaters and waste; resource recovery; providing adequate drainage of urban and rural areas for sanitation; and controlling water quality, soil contamination, and air pollution. The environmental emphasis for the civil engineering degree replaces some of the courses and electives in the general curriculum with further courses in chemistry, biology, and microbiology as well as specific topics in environmental engineering and design.

Student Learning Outcomes

Graduates of the Civil Engineering curriculum should have at the time of graduation:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

PROGRAM EDUCATIONAL OBJECTIVES

By three to five years after graduation, graduates of the civil, construction, and environmental engineering programs will have:

1. Pursued successful careers and expertise in civil engineering, construction engineering, environmental engineering, or a related profession that includes professionalism and increasing responsibility.
2. Collaborated effectively on multi-disciplinary teams to address the needs of society and the environment
3. Pursued lifelong learning, professional development, and licensure as appropriate for their career goals and for developments in the field.

The civil engineering curriculum equips students with a broad education that includes technical skills in analysis and design and professional practices such as communication, teamwork, leadership, and ethics.

The faculty encourages the students to develop their professional skills by participating in cooperative education, internships, or progressive summer engineering employment and study abroad programs. Qualified juniors and seniors interested in graduate studies may apply to the Graduate College to concurrently pursue the bachelor's degree and either a Master of Science in Civil Engineering or a Master of Business Administration in the College of Business Administration. These students would have the opportunity to graduate in five years with both degrees.

Accreditation

The Civil Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>, under the commission's General Criteria and Program Criteria for Civil and Similarly Named Engineering Programs.

Iowa State University is accredited by the Higher Learning Commission (HLC), a commission of the North Central Association of Colleges and Schools.

Explore More

For more information on the undergraduate major in Civil Engineering, please visit:

- <https://www.engineering.iastate.edu/undergraduate-majors-and-minors/civil/>

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College Website

- <https://www.engineering.iastate.edu/>

Department Website

- [ccee.iastate.edu](http://www.ccee.iastate.edu/) (<http://www.ccee.iastate.edu/>)

Program Administration

- **Owning Academic Unit/College:** College of Engineering
- **Coordinating Academic Unit/Department:** Department of Civil, Construction and Environmental Engineering

Credential Opportunities

- **Program Duration:** 4 years
- **Credential(s) Offered:** Civil Engineering, B.S. (129 credits)
- **Program Focus Areas:**
 - **Environmental Engineering Undergraduate Program Focus:**
This program focus is concerned with protecting the public and natural health; providing an ample safe water supply; managing solid and hazardous waste; treating and disposing of domestic and industrial wastewaters and waste; resource recovery; providing adequate drainage of urban and rural areas for sanitation; and controlling water quality, soil contamination, and air pollution. The environmental emphasis for the civil engineering degree replaces some of the courses and electives in the general curriculum with further courses in chemistry, biology, and microbiology as well as specific topics in environmental engineering and design. (132 credits)

- **Available as an Additional Major:** The Civil Engineering, B.S. program of study is not offered as an additional major. It is offered only as a full degree program. Students pursuing this degree program are allowed to pursue additional majors in other eligible programs of study if desired.
- **Program Limitations:** There are no additional program limitations beyond the rules regarding degree planning (<https://catalog.iastate.edu/academics/#degreeplanningtext>).
- **Program Designation:** On-Campus Program

Curriculum in Civil Engineering (General)

Administered by the Department of Civil, Construction and Environmental Engineering.

Leading to the degree bachelor of science.

Total credits required: 128-129.

Any transfer credit courses applied to the degree program require a grade of C or better (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). See also Basic Program and Special Programs. Note: Department does not allow Pass/Not Pass credits to be used to meet graduation requirements.

International Perspectives: 3 cr. ¹

U.S. Cultures and Communities: 3 cr. ¹

Communication Proficiency/Library requirement

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1

Social Sciences and Humanities: 12 cr. ²

Complete 12 cr. with 6 cr. at 2000-level or above.

Basic Program: 24 cr. ³

A minimum GPA of 2.00 required for this set of courses (please note that transfer course grades will not be calculated into the Basic Program GPA). See Requirement for Entry into Professional Program in College of Engineering Overview section.

CHEM 1670	General Chemistry for Engineering Students	4
or CHEM 1770 General Chemistry I & CHEM 1780 and General Chemistry II		
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGR 1010	Engineering Orientation	R
CE 1600	Systematic Problem Solving and Computer Applications for Civil, Construction, and Environmental Engineers ³	3

LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
Total Credits		24

Math and Physical Science: 18-20 cr.

CHEM 1670L	Laboratory in General Chemistry for Engineering	1
	or CHEM 1770LLaboratory in General Chemistry I	
PHYS 2320 & 2320L	Introduction to Classical Physics II and Introduction to Classical Physics II Laboratory	4-5
	or CHEM 1780 General Chemistry II & 1780L and Laboratory in College Chemistry II	
GEOL 2010	Geology for Engineers and Environmental Scientists	3
MATH 2650	Calculus III	4
MATH 2660	Elementary Differential Equations	3
STAT 3005	Engineering Statistics	3-4
	or STAT 3031 Probability and Statistical Inference for Engineers	
Total Credits		18-20

Civil Engineering Core: 34 cr. Minimum GPA of 2.00 required for this set of courses to graduate (including transfer courses; please note that transfer course grades will not be calculated into the Core GPA).

A minimum of 25 credits within the CE-general core must be taken at Iowa State University.

CE 2710 & CE 2720	Engineering Foundations of Statics and Applied Engineering Statics	3
	or CE 2740 Engineering Statics	
EM 3240	Mechanics of Materials	3
ME 3450	Engineering Dynamics	3
CE 2060	Engineering Economic Analysis and Professional Issues in Civil Engineering	3
CE 3260	Principles of Environmental Engineering	3
ABE 3780	Mechanics of Fluids	3
CE 3320	Structural Analysis I	3
CE 3550	Principles of Transportation Engineering	3
CE 3600	Geotechnical Engineering	4
CE 3720	Engineering Hydrology and Hydraulics	3
CE 3820	Design of Concretes	3
Total Credits		34

Other Remaining Courses: 40 cr.

CE 1200	Civil Engineering Learning Community	1
CE 1110	Geomatics	3
CE 1700	Graphics for Civil Engineering	2
CE 3060	Project Management for Civil and Environmental Engineers	3
Any two of the following three courses:		6
CE 3330	Structural Steel Design I	
CE 3340	Reinforced Concrete Design I	
CE 4600	Foundation Engineering	
CE 4850	Civil Engineering Design	3
EM 3270	Mechanics of Materials Laboratory	1
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
SPCM 2120	Fundamentals of Public Speaking	3
Civil Engineering Design Elective ²		3
Technical Communication Elective ²		3
Engineering Topics Electives ²		9
Total Credits		40

Seminar/Co-op/Internships: R cr.

CE 4030	Program and Outcome Assessment	R
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Notes.

1. These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program. U.S. Cultures and Communities and International Perspectives courses may not be taken Pass/Not Pass.
2. Choose from department approved list (<https://www.ccee.iastate.edu/undergraduate-programs/civil-engineering/civil-engineering-general-option-forms-and-information-by-cohort/>). At least six of nine credits must be CE or CONE courses for the Engineering Topics Electives.
3. See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.
4. Students who transfer in with CHEM 1670/CHEM 1670L will be able to take CHEM 1780/CHEM 1780L to complete the program's Chemistry requirement.

See also: A 4-year plan of study grid showing course template by semester for Civil Engineering

Curriculum in Civil Engineering with Environmental Option

Administered by the Department of Civil, Construction and Environmental Engineering.

Leading to the degree Bachelor of Science.

Total credits required: 132.

Any transfer credit courses applied to the degree program require a grade of C or better (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). See also Basic Program and Special Programs.

International Perspectives: 3 cr. ¹

U.S. Cultures and Communities: 3 cr. ¹

Communication Proficiency/Library requirement:

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1

Social Sciences and Humanities: 12 cr. ²

Complete 12 cr. with 6 cr. at 2000-level or above.

Basic Program: 24 cr. ³. Minimum GPA of 2.00 required for this set of courses to graduate, including any transfer courses (please note that transfer course grades will not be calculated into the Basic Program GPA).

CHEM 1770	General Chemistry I	4
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGR 1010	Engineering Orientation	R
CE 1600	Systematic Problem Solving and Computer Applications for Civil, Construction, and Environmental Engineers ³	3
LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I (Minimum Grade of C- if taken at ISU, if transferred in minimum grade must be at least a "C")	4
MATH 1660	Calculus II (Minimum Grade of C- if taken at ISU, if transferred in minimum grade must be at least a "C")	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
Total Credits		24

Math and Physical Science: 27-28 cr.

CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II ⁴	3

CHEM 1780L	Laboratory in College Chemistry II ⁴	1
CHEM 2310	Elementary Organic Chemistry	3
CHEM 2310L	Laboratory in Elementary Organic Chemistry	1
BIOL 2510	Biological Processes in the Environment	3
GEOL 2010	Geology for Engineers and Environmental Scientists	3
MATH 2650	Calculus III	4
MATH 2660	Elementary Differential Equations	3
MICRO 2010	Introduction to Microbiology	2
STAT 3005	Engineering Statistics	3-4
or STAT 3031	Probability and Statistical Inference for Engineers	
Total Credits		27-28

Civil/Env Engineering Core: 31 cr. Minimum GPA of 2.00 required for this set of courses to graduate (including transfer courses; please note that transfer course grades will not be calculated into the Core GPA).

A minimum of 22 credits within the CE-Environmental core must be taken at Iowa State University.

CE 2710	Engineering Foundations of Statics	3
& CE 2720	and Applied Engineering Statics	
or CE 2740	Engineering Statics	
EM 3240	Mechanics of Materials	3
CE 2060	Engineering Economic Analysis and Professional Issues in Civil Engineering	3
CE 3260	Principles of Environmental Engineering	3
CE 3320	Structural Analysis I	3
ABE 3780	Mechanics of Fluids	3
CE 3550	Principles of Transportation Engineering	3
CE 3600	Geotechnical Engineering	4
CE 3720	Engineering Hydrology and Hydraulics	3
CE 3820	Design of Concretes	3
Total Credits		31

Other Remaining Courses: 38 cr.

CE 1200	Civil Engineering Learning Community	1
CE 1110	Geomatics	3
CE 1700	Graphics for Civil Engineering	2
CE 3060	Project Management for Civil and Environmental Engineers	3
CE 3340	Reinforced Concrete Design I	3
CE 4280	Water and Wastewater Treatment Plant Design	3
Any two of the following four courses		6
ENVE 4260	Environmental Engineering Science	
ENVE 4270	Environmental Engineering Systems	

ENVE 4290	Air Pollution and Control	
ENVE 4300	Solid and Hazardous Waste Management	
CE 4850	Civil Engineering Design	3
EM 3270	Mechanics of Materials Laboratory	1
ABE 3780L	Mechanics of Fluids Laboratory	1
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
SPCM 2120	Fundamentals of Public Speaking	3
Civil Engineering Design Elective ²		3
Technical Communication Elective ²		3
Total Credits		38

Seminar/Co-op/Internships: R cr.

CE 4030	Program and Outcome Assessment	R
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Co-op/Internship optional.**Notes.**

1. These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program. U.S. Cultures and Communities and International Perspectives courses may not be taken Pass/Not Pass.
2. Choose from department approved list. (<https://www.ccee.iastate.edu/undergraduate-programs/civil-engineering/civil-engineering-environmental-option-forms-and-information-by-cohort/>)
3. See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.
4. Students who transfer in with CHEM 1670 General Chemistry for Engineering Students/CHEM 1670L Laboratory in General Chemistry for Engineering will be able to take CHEM 1780 General Chemistry II/CHEM 1780L Laboratory in College Chemistry II to complete the program's Chemistry requirement.

See also: A 4-year plan of study grid showing course template by semester for Civil Engineering

Civil Engineering, B.S. - GENERAL Program

Sample Plan - Any four year plans shared in the catalog are samples of a potential university experience.

First Year

Fall	Credits Spring	Credits
MATH 1650	4 MATH 1660	4
ENGR 1010	R PHYS 2310	4
CE 1200	1 PHYS 2310L	1
CE 1600	3 CE 2710	1

CHEM 1670	4 CE 1700	2
CHEM 1670L	1 ENGL 2500	3
ENGL 1500	3 SPCM 2120	3
LIB 1600	1	
	17	18

Second Year

Fall	Credits Spring	Credits
CE 2720	2 EM 3240	3
GEOL 2010	3 ABE 3780	3
MATH 2650	4 CE 1110	3
PHYS 2320	4 CE 3060	3
PHYS 2320L	1 MATH 2660	3
	STAT 3005	3
	14	18

Third Year

Fall	Credits Spring	Credits
CE 3320	3 CE 3340 (CE 4600 may be substituted for CE 3330 or CE 3340)	3
EM 3270	1 CE 3550	3
CE 2060	3 CE 3720	3
CE 3600	4 CE 3820	3
ME 3450	3 Engr Topic Elective	3
Technical Communication Elective	3 SSH Elective	3
	17	18

Fourth Year

Fall	Credits Spring	Credits
CE 3330 (CE 4600 may be substituted for CE 3330 or CE 3340)	3 CE 4850	3
CE 3260	3 CE 4030	R
Engr Topic Elective	3 CE Design Elective	3
International Perspective	3 Engr Topic Elective	3
US Cultures and Communities	3 SSH Elective	3
	15	12

Civil Engineering, B.S. - environmental OPTION

Sample Plan - Any four year plans shared in the catalog are samples of a potential university experience.

First Year			US Cultures and Communities	3 SSH Elective	3
Fall	Credits Spring	Credits			
MATH 1650	4 MATH 1660	4		16	12
ENGR 1010	R PHYS 2310	4	A concurrent bachelor of science/master of science (B.S./M.S.) degree program is available to qualified seniors at Iowa State University. Go to: https://www.ccee.iastate.edu/prospective-graduate-students/concurrent-programs/ for more information. Graduate Study The Department of Civil, Construction and Environmental Engineering offers graduate programs for the degrees of Master of Engineering, Master of Science, and Doctor of Philosophy with a major in civil engineering with areas of specialization in structural engineering, environmental engineering, construction engineering and management, geotechnical engineering, civil engineering materials, transportation engineering, and intelligent infrastructure engineering. The department also offers graduate minors of 12 to 15 credits of coursework to students from other engineering departments. Candidates for the degrees of Master of Engineering and Master of Science are required to complete a total of 30 acceptable graduate credits. The Master of Engineering degree involves all course work. The Master of Science degree requires the preparation of a thesis or creative component. Candidates for the Doctor of Philosophy degree is required to complete a minimum of 72 acceptable graduate credits. Normal prerequisite for graduate study in civil engineering is the completion of an undergraduate curriculum substantially equivalent to that required of engineering students at this university. Because of the diversity of interests within the graduate programs in civil engineering, a student may qualify for graduate study even though undergraduate or prior graduate training has been in a discipline other than engineering. Additional courses will be required depending upon the student's background and area of interest. The department participates in the interdepartmental graduate programs in environmental science and wind energy science, engineering and policy. The department also offers graduate certificates in construction management, environmental engineering, and environmental systems. The construction management certificate requires 12 graduate credits (nine credits of core courses and three credits of approved elective courses). The environmental engineering and environmental systems certificates require 12 graduate credits (six credits of "core courses", six credits of approved "elective courses") and a seminar course or an approved equivalent. Additional information about graduate programs, specialization areas, and admission criteria are available on the department's website http://		
CE 1200	1 PHYS 2310L	1			
CE 1600	3 CE 2710	1			
CHEM 1770	4 CE 1700	2			
CHEM 1770L	1 ENGL 2500	3			
ENGL 1500	3 BIOL 2510 (Spring only)	3			
LIB 1600	1				
	17	18			
Second Year					
Fall	Credits Spring	Credits			
CE 2720	2 EM 3240	3			
GEOL 2010	3 CE 2060	3			
MATH 2650	4 CE 1110	3			
CHEM 1780	3 CE 3060	3			
CHEM 1780L	1 MATH 2660	3			
SSH Elective	3 STAT 3005	3			
	16	18			
Third Year					
Fall	Credits Spring	Credits			
CE 3320	3 CE 3340	3			
EM 3270	1 CE 3260	3			
CE 3550	3 MICRO 2010	2			
CE 3820	3 CE 3720	3			
ABE 3780	3 SPCM 2120	3			
ABE 3780L	1 Technical Communication Elective	3			
CHEM 2310	3				
CHEM 2310L	1				
	18	17			
Fourth Year					
Fall	Credits Spring	Credits			
CE 3600	4 CE 4850	3			
Take two of four: ENVE 4260, 4270, 4290 or 4300	3 CE 4280	3			
ENVE 4260 (Fall only)	Take two of four: ENVE 4260, 4270, 4290 or 4300	3			
ENVE 4300 (Fall only)	ENVE 4270 (Spring only)				
CE Design Elective	3 ENVE 4290 (Spring only)				
International Perspective	3 CE 4030	R			

www.ccee.iastate.edu/ccee-advanced-degree-programs (<https://www.ccee.iastate.edu/ccee-advanced-degree-programs/>).